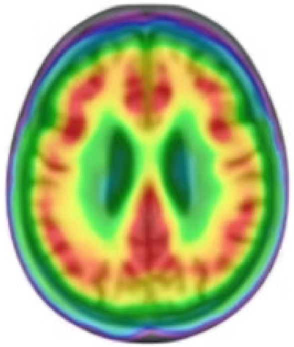
Deep Learning for Alzheimer's Disease Diagnosis

Using advanced neuroimaging data and deep learning architectures to classify Alzheimer's disease stages with unprecedented accuracy.



The Challenge of Alzheimer's Diagnosis

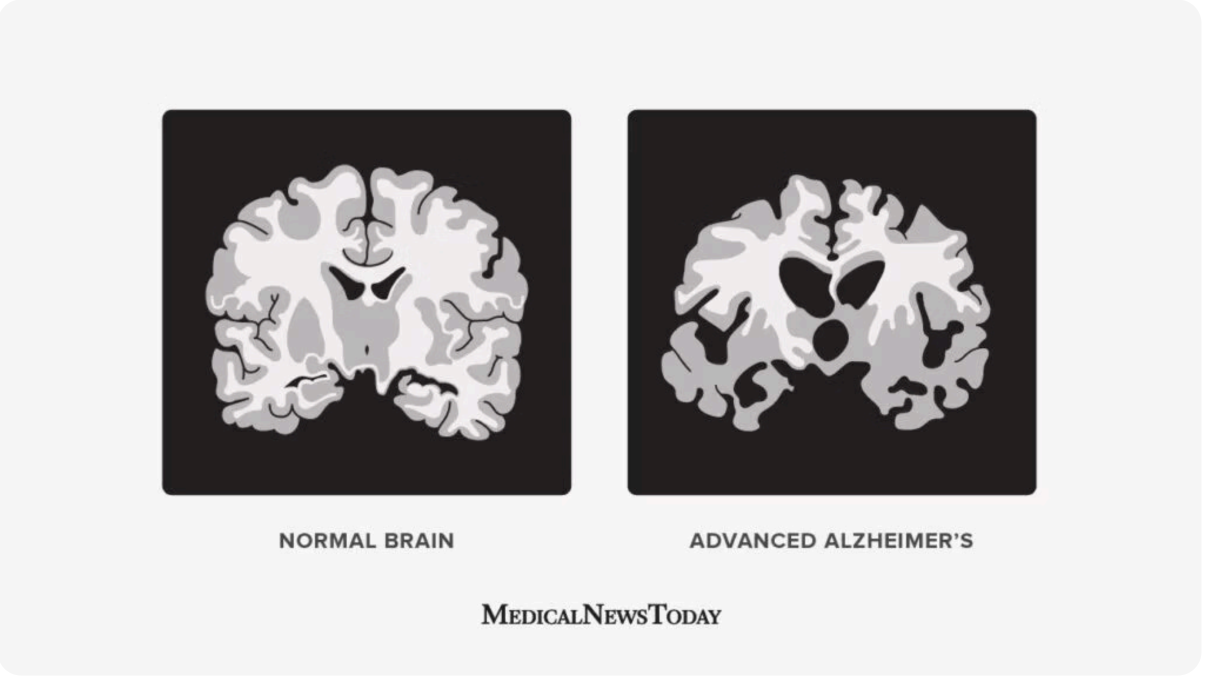
The Problem

Alzheimer's disease causes gradual decline in memory, thinking ability, and daily functioning due to protein accumulation in the brain. Early identification through MRI is crucial for diagnosis and monitoring disease progression.

Patients are classified into three major stages: **CN** (Cognitively Normal), **MCI** (Mild Cognitive Impairment), and **AD** (Alzheimer's Disease).

Structural Changes

- Hippocampus shrinkage and severe atrophy
- Cortical thinning in multiple brain regions
- Enlargement of brain ventricles
- Degeneration of entorhinal cortex
- Reduced metabolism in posterior cingulate



Dataset & Research Foundation

ADNI Dataset

18,030 MRI slices across four classes: AD, CN, EMCI, and MCI. Approximately 15 slices per patient, resulting in 282-308 patients per class.

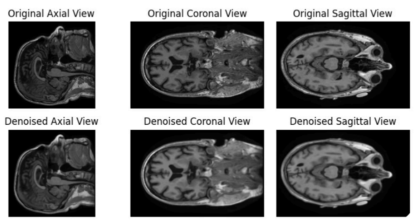
Balanced Distribution

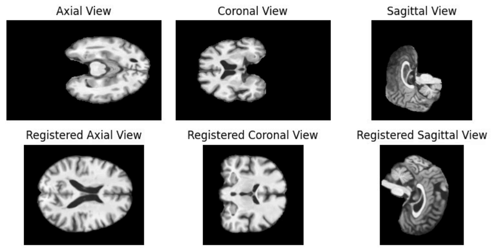
Dataset carefully balanced across all diagnostic categories to ensure fair model training and evaluation.

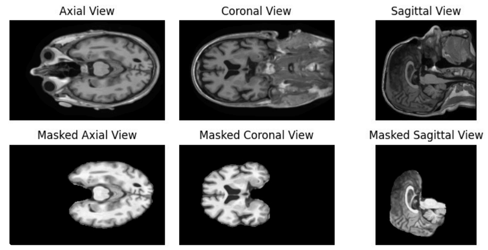
Multi-View Approach

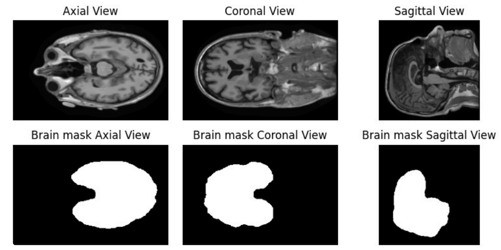
15 middle slices extracted from each anatomical view (Sagittal, Coronal, Axial) to increase dataset size and model generalizability.

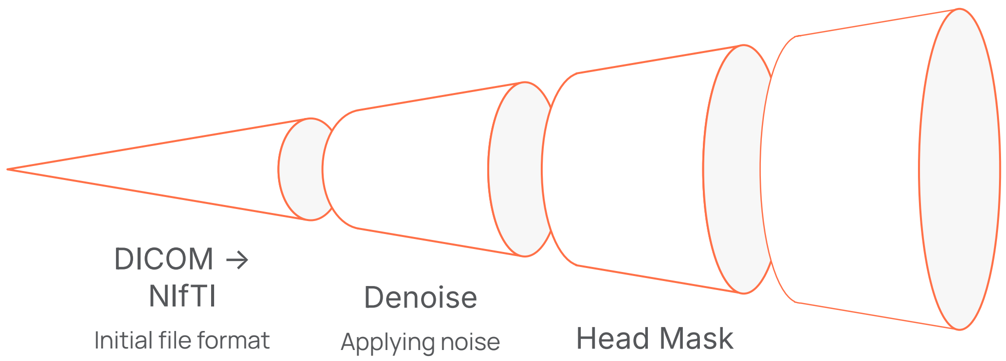
Data Preprocessing Pipeline











DICOM → NIfTI

Initial file format conversion for processing.

Denoise

Applying noise reduction algorithms to improve image quality.

Head Mask

Isolating the region of interest for analysis.

Bias Correction

Adjusting for field inhomogeneities in the data.

Our comprehensive preprocessing pipeline transforms raw DICOM files into standardized, analysis-ready MRI data through multiple stages of refinement and normalization.

01

Format Conversion & Validation

DICOM files converted to SimpleITK format with IAL orientation and Float32 datatype for numerical stability.

02

Denoising & Masking

Head masks created through intensity rescaling and thresholding to isolate brain tissue.

03

Bias Field Correction

N4BiasFieldCorrectionImageFilter applied to remove intensity non-uniformities across the scan.

04

Brain Extraction

ANTsPyNet library used to create precise brain masks and extract brain regions.

05

Template Registration

SyN algorithm aligns scans with MNI-152 template for standardization across subjects.

06

Final Normalization

Z-normalization, resampling to 1mm spacing, resizing to (224,224,224) voxels, and intensity rescaling to [-1,1].

Model Architecture Comparison

We evaluated six distinct architectures ranging from shallow custom CNNs to deep pretrained models, each offering different trade-offs between complexity, parameter efficiency, and representational power.

CNN-5Layer

Baseline model with 6 conv layers, 2M parameters. Filter progression: 32→64→128→256→512→656. Dropout: 0.33.

CNN-10Layer

Medium depth with double convolutions per block, 8M parameters. Dense units: 1024. Dropout: 0.28.

CNN-15Layer

Deep custom CNN with 3 convolutions per stage, 15M parameters. Maximum feature extraction capability.

ResNet50

Transfer learning with 50 layers, 25M total parameters. Residual connections stabilize deep training.

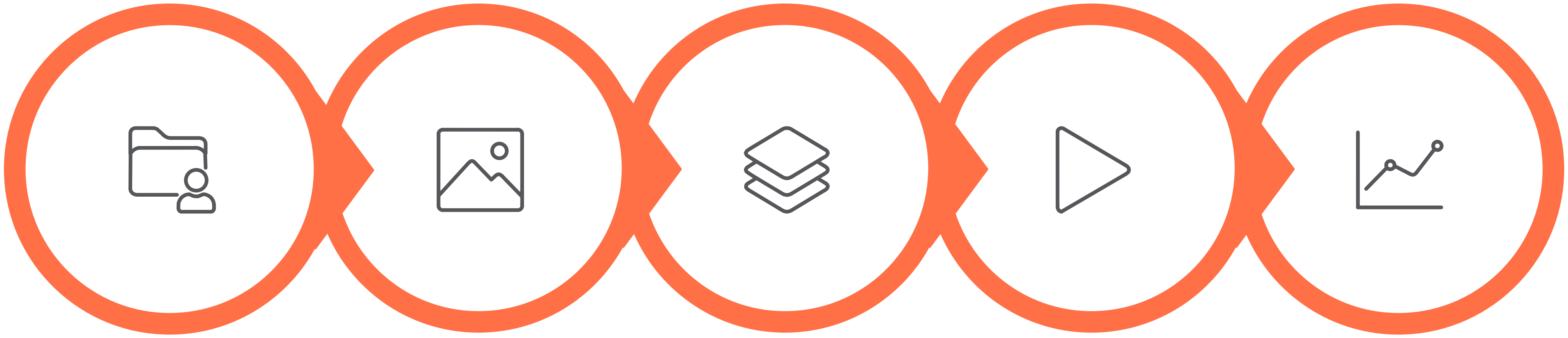
ResNet101

Deepest backbone with 101 layers, 44M parameters. 23 additional blocks in Stage 3 for superior feature extraction.

EfficientNet-B3

Efficient backbone with SE modules, 12M parameters. Best efficiency-accuracy trade-off with attention mechanisms.

Training Methodology & Workflow



Dataset Setup

Data Augmentatio n

Model Selection

Training w/ Callbacks

Evaluation

Data Augmentation Strategy

Conservative augmentation techniques applied to preserve anatomical brain structure while increasing dataset robustness:

- **Rotation:** ±8 degrees to simulate scan misalignment
- **Spatial shifts:** 4% height/width range
- **Brightness variation:** Simulating scanner differences
- **Fill mode:** Nearest to avoid artificial edges

Training Configuration

Class weights calculated to address imbalance. Models trained with Adam optimizer, categorical crossentropy loss, and early stopping based on validation AUC.

224

Input Size

224×224×3 standardized input dimensions

200

Max Epochs

With early stopping patience

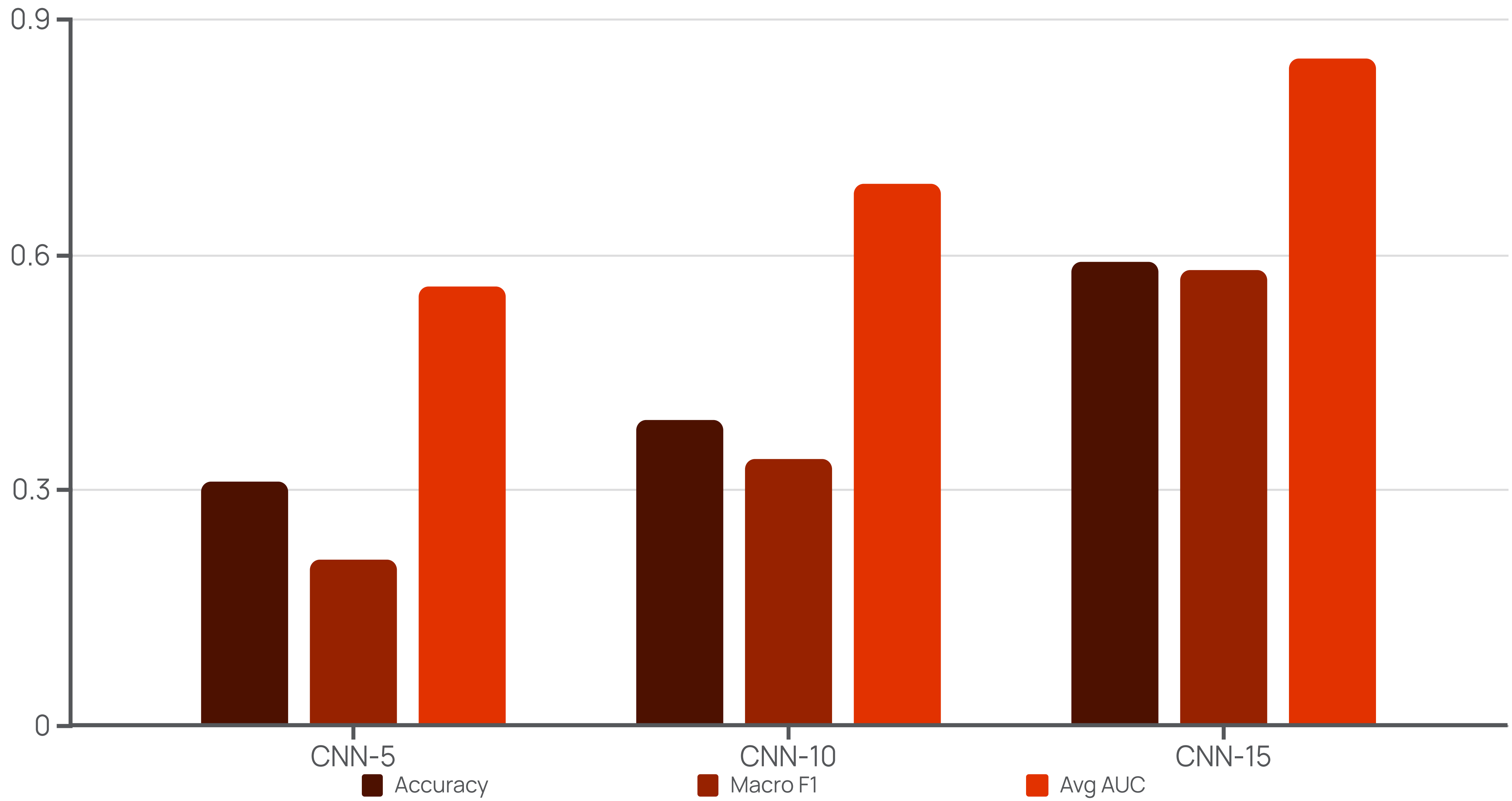
4

Classes

AD, MCI, CN, EMCI

Performance Results: Custom CNNs

Custom CNN architectures showed progressive improvement with depth, but struggled with class separability compared to pretrained models.



1

CNN-5 Baseline

Limited discriminative capability with 31% accuracy. Strong bias toward CN class. Underfitting evident in training curves.

2

CNN-10 Improvement

Double convolutions per stage improved accuracy to 39%. Better EMCI recognition (70% recall) but CN remained challenging.

3

CNN-15 Best Custom

Three convolutions per stage achieved 59% accuracy. Strong multi-scale feature extraction with 0.85 AUC and excellent EMCI detection (86%).

Transfer Learning Excellence

Parameter Efficiency

EfficientNet-B3 achieved competitive results with fewer parameters than ResNet variants through compound scaling strategy.

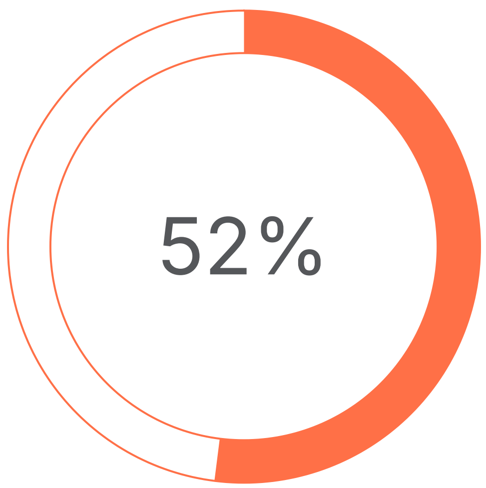
Class Balance

Transfer learning models showed more balanced per-class performance compared to custom CNNs, reducing bias toward majority classes.

Generalization

Pretrained ImageNet weights provided strong feature extraction foundation, significantly reducing overfitting on medical imaging tasks.

EfficientNet-B3 Results



Accuracy

Balanced performance across all classes

Micro-Avg AUC

Strong class separability

EfficientNet-B3 delivered the best efficiency-accuracy trade-off with only 12M parameters. Squeeze-and-excitation modules effectively emphasized relevant MRI channels.

ResNet50 Performance

Achieved 50% accuracy with strong AD separation (AUC 0.83). Residual connections enabled stable deep training with reasonable generalization.

ResNet101

91% Accuracy

ResNet101 achieved breakthrough performance, demonstrating the power of deep residual architectures for medical image classification.

0.91
Overall Accuracy

Across all four diagnostic classes

0.99
Average AUC

Near-perfect class separability

0.91
Macro F1-Score

Balanced precision and recall

44M
Parameters

Deep architecture capacity

AD Classification

89% recall with 96% precision. Strong identification of advanced Alzheimer's cases.

MCI Detection

91% recall with 92% precision. Excellent mild cognitive impairment recognition.

CN Recognition

92% recall with 87% precision. Accurate identification of cognitively normal patients.

EMCI Identification

92% recall with 91% precision. Superior early-stage detection capability.

Web Application & Inference Pipeline

To translate research into practical clinical utility, we developed an end-to-end web application using Streamlit that integrates neuroimaging preprocessing, deep learning inference, and an intuitive user interface for real-time Alzheimer's stage prediction from brain MRI scans.

Technology Stack

- Frontend: Streamlit (Python-based web UI)
- Deep Learning: TensorFlow/Keras with ResNet101
- Neuroimaging: SimpleITK, ANTsPy, TorchIO, NiBabel, ANTsPyNet
- Processing: NumPy, OpenCV, PIL, Matplotlib

Preprocessing Pipeline

Automated 8-step pipeline ensures consistent MRI normalization

- Orientation standardization and denoising
- N4 bias field correction with head mask
- Brain extraction (skull stripping)
- Non-linear registration to MNI152 template
- Resampling to 1mm isotropic resolution
- Z-score normalization and intensity rescaling

Inference Workflow

MRI Upload - User uploads NIfTI file (.nii/.nii.gz)

Slice Extraction - Extract 15 middle axial slices at 224×224

ResNet101 Prediction - Per-slice classification with softmax

Gaussian Aggregation - Weighted average centered on middle slices

Results Display - Predicted class (AD/MCI/CN) with confidence score

Demo Link: https://drive.google.com/file/d/11FbzjjolPn-an45nIRNFYrEzmSvx5GR2/view?usp=sharing

Research Tool Only: This application demonstrates the clinical deployment potential of our ResNet101 model. The system processes a full 3D MRI scan in under 2 minutes, providing probability distributions across all diagnostic classes with visual slice previews.

Conclusions & Future Directions

Key Findings

This study demonstrates that deep pretrained architectures are essential for reliable medical image classification. ResNet101 consistently outperformed all other models, achieving 91% accuracy with near-perfect AUC scores across all Alzheimer's stages.

The performance gap between custom CNNs (31-59% accuracy) and transfer learning models (50-91% accuracy) underscores the value of leveraging large-scale pretrained weights for medical imaging tasks.

Future Work: Multi-Modal Integration

Our next phase will extend this framework into a **comprehensive multi-modal diagnostic model** by integrating:

- Genetic biomarkers (APOE-ε4 status)
- Cognitive assessment scores
- Clinical questionnaires and demographics
- Biochemical markers

Data Architecture

Design multi-modal fusion pipeline

Hybrid Models

Transformer-based fusion networks

External Validation

Multi-site testing and robustness analysis

Clinical Deployment

Real-world healthcare integration

Clinical Impact: This research demonstrates promising potential for reliable automatic Alzheimer's stage classification. However, external validation on diverse cohorts and cross-site testing are essential before clinical deployment.

References

This research builds upon recent advances in deep learning for Alzheimer's disease diagnosis and neuroimaging analysis.

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